

Customer Support Automation System

Installing dependencies and setting up environment:

```
python -m venv venv
```

```
venv\Scripts\activate
```

```
python -m pip install --upgrade pip
```

```
pip install python-dotenv langchain-google-genai langchain-core langchain-community langchain-text-splitters langgraph faiss-cpu
```

OR

Create requirements.txt and then include:

```
python-dotenv
```

```
langchain-google-genai
```

```
langchain-core
```

```
langchain-community
```

```
langchain-text-splitters
```

```
langgraph
```

```
faiss-cpu
```

Then run : `pip install -r requirements.txt`

We then write the code and run : `python main.py`

```

PS C:\Users\prash\Downloads\ASSIGNMENT2_CUSTOMER_SUPPORT_AI_AkankshaKulkarni_24BCE1818(akanksha.prashant2024@vitstudent.ac.in)\src> python -m venv venv
PS C:\Users\prash\Downloads\ASSIGNMENT2_CUSTOMER_SUPPORT_AI_AkankshaKulkarni_24BCE1818(akanksha.prashant2024@vitstudent.ac.in)\src> venv\Scripts\activate
(venv) PS C:\Users\prash\Downloads\ASSIGNMENT2_CUSTOMER_SUPPORT_AI_AkankshaKulkarni_24BCE1818(akanksha.prashant2024@vitstudent.ac.in)\src> python -m pip install
--upgrade pip
Requirement already satisfied: pip in c:\users\prash\downloads\assignment2_customer_support_ai_akankshakulkarni_24bce1818(akanksha.prashant2024@vitstudent.ac.in)
)\src\venv\lib\site-packages (24.3.1)
Collecting pip
  Downloading pip-26.1.2-py3-none-any.whl.metadata (4.6 kB)
  Downloading pip-26.1.2-py3-none-any.whl (1.8 MB)
    1.8/1.8 MB 841.7 kB/s eta 0:00:00
Installing collected packages: pip
  Attempting uninstall: pip
    Found existing installation: pip 24.3.1
    Uninstalling pip-24.3.1:
      Successfully uninstalled pip-24.3.1
Successfully installed pip-26.1.2
(venv) PS C:\Users\prash\Downloads\ASSIGNMENT2_CUSTOMER_SUPPORT_AI_AkankshaKulkarni_24BCE1818(akanksha.prashant2024@vitstudent.ac.in)\src> pip install python-do
tenv langchain-google-genai langchain-core langchain-community langchain-text-splitters langgraph faiss-cpu
Collecting python-dotenv
  Using cached python_dotenv-1.2.2-py3-none-any.whl.metadata (27 kB)
Collecting langchain-google-genai
  Using cached langchain_google_genai-4.2.6-py3-none-any.whl.metadata (2.7 kB)
Collecting langchain-core
  Using cached langchain_core-1.4.8-py3-none-any.whl.metadata (4.7 kB)
Collecting langchain-community
  Using cached langchain_community-0.4.2-py3-none-any.whl.metadata (3.4 kB)
Collecting langchain-text-splitters
  Using cached langchain_text_splitters-1.1.2-py3-none-any.whl.metadata (3.3 kB)
Collecting langgraph
  Using cached langgraph-1.2.6-py3-none-any.whl.metadata (4.9 kB)
Collecting faiss-cpu
  Using cached faiss_cpu-1.14.3-cp313-cp313-win_amd64.whl.metadata (7.8 kB)
Collecting filetype<2.0.0,>=1.2.0 (from langchain-google-genai)
  Using cached filetype-1.2.0-py2.py3-none-any.whl.metadata (6.5 kB)
Collecting google-genai<3.0.0,>=1.65.0 (from langchain-google-genai)
  Using cached google_genai-2.10.0-py3-none-any.whl.metadata (53 kB)
Collecting pydantic<3.0.0,>=2.0.0 (from langchain-google-genai)
  Using cached pydantic-2.13.4-py3-none-any.whl.metadata (109 kB)
Collecting jsonpatch<2.0.0,>=1.33.0 (from langchain-core)
  Using cached jsonpatch-1.33.0-py3-none-any.whl.metadata (3.0 kB)

```

```

(venv) PS C:\Users\prash\Downloads\ASSIGNMENT2_CUSTOMER_SUPPORT_AI_AkankshaKulkarni_24BCE1818(akanksha.prashant2024@vitstudent.ac.in)\src> pip install python-do
tenv langchain-google-genai langchain-core langchain-community langchain-text-splitters langgraph faiss-cpu
Requirement already satisfied: python-dotenv in .\venv\lib\site-packages (1.2.2)
Requirement already satisfied: langchain-google-genai in .\venv\lib\site-packages (4.2.6)
Requirement already satisfied: langchain-core in .\venv\lib\site-packages (1.4.8)
Requirement already satisfied: langchain-community in .\venv\lib\site-packages (0.4.2)
Requirement already satisfied: langchain-text-splitters in .\venv\lib\site-packages (1.1.2)
Requirement already satisfied: langgraph in .\venv\lib\site-packages (1.2.6)
Requirement already satisfied: faiss-cpu in .\venv\lib\site-packages (1.14.3)
Requirement already satisfied: filetype<2.0.0,>=1.2.0 in .\venv\lib\site-packages (from langchain-google-genai) (1.2.0)
Requirement already satisfied: google-genai<3.0.0,>=1.65.0 in .\venv\lib\site-packages (from langchain-google-genai) (2.10.0)
Requirement already satisfied: pydantic<3.0.0,>=2.0.0 in .\venv\lib\site-packages (from langchain-google-genai) (2.13.4)
Requirement already satisfied: jsonpatch<2.0.0,>=1.33.0 in .\venv\lib\site-packages (from langchain-core) (1.33)
Requirement already satisfied: langchain-protocol<0.0.17 in .\venv\lib\site-packages (from langchain-core) (0.0.18)
Requirement already satisfied: langsmith<1.0.0,>=0.3.45 in .\venv\lib\site-packages (from langchain-core) (0.9.2)
Requirement already satisfied: packaging<23.2.0 in .\venv\lib\site-packages (from langchain-core) (26.2)
Requirement already satisfied: pyyaml<7.0.0,>=5.3.0 in .\venv\lib\site-packages (from langchain-core) (6.0.3)
Requirement already satisfied: tenacity<8.4.0,<10.0.0,>=8.1.0 in .\venv\lib\site-packages (from langchain-core) (9.1.4)
Requirement already satisfied: typing-extensions<5.0.0,>=4.7.0 in .\venv\lib\site-packages (from langchain-core) (4.15.0)
Requirement already satisfied: uuid-utils<1.0,>=0.12.0 in .\venv\lib\site-packages (from langchain-core) (0.16.2)
Requirement already satisfied: anyio<5.0.0,>=4.8.0 in .\venv\lib\site-packages (from google-genai<3.0.0,>=1.65.0->langchain-google-genai) (4.14.1)
Requirement already satisfied: google-auth<3.0.0,>=2.48.1 in .\venv\lib\site-packages (from google-auth[requests]<3.0.0,>=2.48.1->google-genai<3.0.0,>=1.65.0->langchain-google-genai) (2.55.1)
Requirement already satisfied: httpx<1.0.0,>=0.28.1 in .\venv\lib\site-packages (from google-genai<3.0.0,>=1.65.0->langchain-google-genai) (0.28.1)
Requirement already satisfied: requests<3.0.0,>=2.28.1 in .\venv\lib\site-packages (from google-genai<3.0.0,>=1.65.0->langchain-google-genai) (2.34.2)
Requirement already satisfied: websockets<17.0,>=13.0.0 in .\venv\lib\site-packages (from google-genai<3.0.0,>=1.65.0->langchain-google-genai) (15.0.1)
Requirement already satisfied: distro<2,>=1.7.0 in .\venv\lib\site-packages (from google-genai<3.0.0,>=1.65.0->langchain-google-genai) (1.9.0)
Requirement already satisfied: sniffio in .\venv\lib\site-packages (from google-genai<3.0.0,>=1.65.0->langchain-google-genai) (1.3.1)
Requirement already satisfied: idna<=2.8 in .\venv\lib\site-packages (from anyio<5.0.0,>=4.8.0->google-genai<3.0.0,>=1.65.0->langchain-google-genai) (3.18)
Requirement already satisfied: pyasn1-modules<0.2.1 in .\venv\lib\site-packages (from google-auth<3.0.0,>=2.48.1->google-auth[requests]<3.0.0,>=2.48.1->google-genai<3.0.0,>=1.65.0->langchain-google-genai) (0.4.2)
Requirement already satisfied: cryptography<=38.0.3 in .\venv\lib\site-packages (from google-auth<3.0.0,>=2.48.1->google-auth[requests]<3.0.0,>=2.48.1->google-genai<3.0.0,>=1.65.0->langchain-google-genai) (49.0.0)
Requirement already satisfied: certifi in .\venv\lib\site-packages (from httpx<1.0.0,>=0.28.1->google-genai<3.0.0,>=1.65.0->langchain-google-genai) (2026.6.17)
Requirement already satisfied: httpcore==1.* in .\venv\lib\site-packages (from httpx<1.0.0,>=0.28.1->google-genai<3.0.0,>=1.65.0->langchain-google-genai) (1.0.9)
Requirement already satisfied: h11<=0.16 in .\venv\lib\site-packages (from httpcore==1.*->httpx<1.0.0,>=0.28.1->google-genai<3.0.0,>=1.65.0->langchain-google-genai) (0.16.0)
Requirement already satisfied: isoparser<1.9 in .\venv\lib\site-packages (from jsonpatch<2.0.0,>=1.33.0->langchain-core) (3.1.1)

```

INITIAL OUTPUT:

```
(venv) PS C:\Users\prash\Downloads\ASSIGNMENT2_CUSTOMER_SUPPORT_AI_AkankshaKulkarni_24BCE1818(akanksha.prashant2024@vitstudent.ac.in)\src> python main.py
C:\Users\prash\Downloads\ASSIGNMENT2_CUSTOMER_SUPPORT_AI_AkankshaKulkarni_24BCE1818(akanksha.prashant2024@vitstudent.ac.in)\src\main.py:15: DeprecationWarning:
'langchain-community' is being sunset and is no longer actively maintained. See https://github.com/langchain-ai/langchain-community/issues/674 for details
and migration guidance toward standalone integration packages.
  from langchain_community.vectorstores import FAISS

=====
ABC Technologies Customer Support AI System
Powered by LangGraph + Gemini + FAISS + SQLite
=====

Gemini API key found.

Initializing LLM (Gemini 2.5 Flash)...

Setting up RAG Pipeline (loading knowledge base documents)...
  Loaded: company_policy.txt
  Loaded: pricing_guide.txt
  Loaded: technical_manual.txt
  Loaded: faq.txt
  Created 26 document chunks for retrieval
  Vector store created successfully!

Initializing SQLite memory manager...
SQLite database initialized: memory.db

Building LangGraph workflow...
Workflow compiled successfully!

=====
Choose mode:
  1 - Run Demo (5 sample queries from assignment)
  2 - Interactive Mode (type your own queries)
=====
Enter choice (1 or 2): 1
```

OUTPUTS OF ALL GIVEN TEST CASES:

Test Case 1: What are the pricing plans available for your software?

```
QUERY 1 OF 5
Customer      : Alice Johnson (ID: CUST_001)
Query         : What are the pricing plans available for your software?
Expected      : Sales
=====

[Intent Classification] Analyzing query: 'What are the pricing plans available for your software?...'
[✓] Intent classified as: Sales

[Router] Routing to: sales_agent

[Sales Agent] Processing query...
[✓] Sales agent response generated

[Supervisor Agent] Reviewing and finalizing response...
[✓] Final response prepared by supervisor

RESULT SUMMARY:
Intent Detected : Sales
Approval Required: False
Approval Status : not_required

FINAL RESPONSE TO CUSTOMER:
-----
Subject: Your ABC Technologies Software Pricing Inquiry

Dear Alice Johnson,

Thank you for your interest in ABC Technologies! I'd be delighted to provide you with a clear overview of our software pricing plans, designed to accommodate a variety of business needs.

We offer three primary plans:

1. **STARTER PLAN**
   * **Pricing:** $29 per month or $290 annually (offering a 17% saving).
   * **Ideal for:** Small teams and individuals.
   * **Key Features:** Supports up to 5 users, includes 10 GB of cloud storage, basic reporting and analytics, email support, standard integrations (like Google Workspace and Slack), and API access for up to 1,000 calls per month.
```

Test Case 2: I forgot my account password.

```
Press ENTER to continue to next query...

=====
QUERY 2 OF 5
Customer   : Bob Smith (ID: CUST_002)
Query      : I forgot my account password and cannot log in. How do I reset it?
Expected   : Account
=====

[Intent Classification] Analyzing query: 'I forgot my account password and cannot log in. How do I res...'
Intent classified as: Account

[Router] Routing to: account_agent

[Account Agent] Processing query...
Account agent response generated

[Supervisor Agent] Reviewing and finalizing response...
Final response prepared by supervisor

RESULT SUMMARY:
Intent Detected   : Account
Approval Required: False
Approval Status  : not_required

FINAL RESPONSE TO CUSTOMER:
-----
Subject: Resetting Your ABC Technologies Account Password

Dear Bob Smith,

Thank you for contacting ABC Technologies Support. I understand you're having trouble logging in because you've forgotten your password, and I apologize for any inconvenience this may cause. I'm here to help you quickly regain access to your account.

Here are the simple steps to reset your password:

1.  **Visit our Login Page:** Please go to the ABC Technologies login page.
2.  **Click "Forgot Password":** On the login page, you will find a "Forgot Password" link, typically located below the login fields. Click on this link.
3.  **Enter Your Email:** You will be prompted to enter the email address associated with your ABC Technologies account. Please enter it and submit.
4.  **Check Your Email:** You will receive a password reset link in your inbox within 5 minutes. Please check your spam or junk folder if you don't see it in
```

Query 3 My application crashes whenever I upload a file

```
=====
QUERY 3 OF 5
Customer   : Carol White (ID: CUST_003)
Query      : My application crashes whenever I try to upload a file. This is very frustrating!
Expected   : Technical Support
=====

[Intent Classification] Analyzing query: 'My application crashes whenever I try to upload a file. This...'
Intent classified as: Technical

[Router] Routing to: technical_agent

[Technical Agent] Processing query...
Technical agent response generated

[Supervisor Agent] Reviewing and finalizing response...
Final response prepared by supervisor

RESULT SUMMARY:
Intent Detected   : Technical
Approval Required: False
Approval Status  : not_required

FINAL RESPONSE TO CUSTOMER:
-----
Subject: Re: Application Crashing During File Upload - We're Here to Help, Carol!

Dear Carol White,

Thank you for reaching out to ABC Technologies. I'm very sorry to hear that your application is crashing when you attempt to upload a file. I understand how frustrating this can be, and please be assured that we're committed to helping you get this resolved quickly.

To help us pinpoint the cause and provide the most effective solution, could you please share a few more details about the issue?

1.  **Browser:** Which web browser are you currently using (e.g., Google Chrome, Mozilla Firefox, Microsoft Edge, Safari)?
2.  **Operating System:** What operating system is your computer running (e.g., Windows 10, macOS Ventura, Linux)?
3.  **Steps to Reproduce:** Could you describe the exact steps you take before the crash occurs? For example, "I click the 'Upload' button, select a file, and then the app crashes."
4.  **File Details:** What is the **size** and **format** of the file you are trying to upload?
```

Query 4 I need a refund for my annual subscription.

```
=====
QUERY 4 OF 5
Customer   : David Brown (ID: CUST_004)
Query      : I need a refund for my annual subscription. I purchased it last week but want to cancel.
Expected   : Billing -> Human Approval Required
=====

[Intent Classification] Analyzing query: 'I need a refund for my annual subscription. I purchased it 1...'
Intent classified as: Billing

[Router] Routing to: billing_agent

[Billing Agent] Processing query...
HIGH-RISK REQUEST DETECTED: Refund Request - Requires human approval!
Billing agent response generated

=====
HUMAN SUPERVISOR APPROVAL REQUIRED
=====
Customer ID      : CUST_004
Customer Name    : David Brown
Request Type     : Refund Request
Customer Query   : I need a refund for my annual subscription. I purchased it last week but want to cancel.

AI Draft Response Preview:
-----
I understand you'd like to cancel your annual subscription and receive a refund, especially since you just purchased it last week. I can certainly help you with this.

According to our company policy, all annual subscription cancellations and refund requests require a review and approval from a human supervisor.

=====
Supervisor Decision - Approve this request? (yes/no): yes
Request APPROVED by supervisor

[Supervisor Agent] Reviewing and finalizing response...
Final response prepared by supervisor
```

Query 5 What was my previous support issue?

```
=====
QUERY 5 OF 5
Customer   : Alice Johnson (ID: CUST_001)
Query      : What was my previous support issue?
Expected   : Memory Recall
=====

[Intent Classification] Analyzing query: 'What was my previous support issue?...'
Intent classified as: Memory

[Router] Routing to: memory_recall_agent

[Memory Recall Agent] Retrieving conversation history...
Memory recall response generated

[Supervisor Agent] Reviewing and finalizing response...
Final response prepared by supervisor

RESULT SUMMARY:
Intent Detected   : Memory
Approval Required : False
Approval Status   : not_required

FINAL RESPONSE TO CUSTOMER:
-----
Dear Alice Johnson,

Thank you for reaching out. I've reviewed your support history for you.

Your previous support issue, and indeed your recent inquiries, have consistently focused on the pricing plans available for ABC Technologies software. Our records indicate this has been the primary topic of your interactions, including those today and on June 25th and 26th.

To ensure we fully address your needs, would you like me to provide you with the most current pricing plan details, or perhaps connect you with a specialist who can walk you through the options and answer any specific questions you might have?

Please let me know how I can best assist you further.
```

INTERACTIVE MODE:

```
=====
INTERACTIVE MODE
Type your customer queries to test the system.
Type 'quit' to exit.
=====

Enter your Customer ID (e.g., CUST_001): CUST_3908
Enter your name: Akanksha

Welcome, Akanksha! How can I help you today?

-----
You: I want to return an item that was damaged but the option says there is no refund on it

[Intent Classification] Analyzing query: 'I want to return an item that was damaged but the option say...'
Intent classified as: Billing

[Router] Routing to: billing_agent

[Billing Agent] Processing query...
HIGH-RISK REQUEST DETECTED: Refund Request - Requires human approval!
Billing agent response generated

=====
HUMAN SUPERVISOR APPROVAL REQUIRED
=====
Customer ID : CUST_3908
Customer Name : Akanksha
Request Type : Refund Request
Customer Query : I want to return an item that was damaged but the option says there is no refund on it

AI Draft Response Preview:
-----
I understand you're looking to return a damaged item and are seeing some conflicting information about refunds. I can certainly help clarify the process for you.

Your request for a refund needs to be reviewed by a human supervisor, as per our company policy. I've flagged your request for their immediate review.

=====
Supervisor Decision - Approve this request? (yes/no): no
Please enter rejection reason: item policy
Request REJECTED by supervisor

[Supervisor Agent] Reviewing and finalizing response...
Final response prepared by supervisor

Support AI [Billing]: Subject: Update Regarding Your Damaged Item Return Request - Case ID: [Insert Case ID if available]

Dear [Customer Name],

Thank you for your patience while we thoroughly reviewed your request to return the damaged item and the refund eligibility.

I understand you were looking to return an item that arrived damaged, and that the system indicated no refund was available. As promised by our AI agent, your request was escalated for a human supervisor review, and I have personally overseen the assessment of your situation.

Upon careful review of your request and in accordance with our current return policy for damaged items, we regret to inform you that we are unable to approve a full refund at this time. This decision is based on the specific details of your case as they align with our established guidelines.

We understand this news may be disappointing, and we genuinely empathize with your frustration regarding the damaged item. While a full refund isn't possible under these circumstances, we are committed to finding an alternative solution to ensure your satisfaction.

We would like to offer you the following options:

1. **Partial Store Credit:** We can offer you a [e.g., 25-50%] store credit towards a future purchase with ABC Technologies as a gesture of goodwill.
2. **Discount on a Replacement:** If you wish to purchase a replacement for the damaged item, we can offer you a significant discount on a new unit.
3. **Further Discussion:** If you believe there is additional information or context that was not considered, or if you would like to discuss these options further, please reply to this email or call us directly at [Your Direct Phone Number, e.g., 1-800-XXX-XXXX, Ext. YYYY] at your earliest convenience.

Should you wish to escalate this matter further, you may contact our Customer Relations Manager, [Manager's Name, e.g., Sarah Chen], directly at [Manager's Email Address, e.g., manager.support@abctechnologies.com]. Please reference your Case ID: [Insert Case ID] in your communication.
```

Submitted By: Akanksha Prashant Kumar Kulkarni